

Shangmin Chen

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EDUCATION

Boston University

Bachelor of Arts in Computer Science

Boston, MA

September 2022 – May 2026

EXPERIENCE

Software Engineer

September 2024 – December 2024

BU Spark!

Boston, MA

- Developed full-stack food waste platform using Django, JavaScript, and SQLite, connecting 500+ students and reducing campus waste by 10%
- Built RESTful API with sub-100ms query times and deployed via CI/CD pipelines on Heroku
- Implemented Auth0 authentication and Django security middleware, ensuring zero security incidents
- Accelerated release by 1 week using Agile methodology in team of 5, contributing to 7 iterations

IT Consultant

January 2024 – July 2025

Boston University Engineering IT

Boston, MA

- Automated software deployment for 100+ devices using Microsoft Deployment Toolkit and created custom USB deployment solution for Windows 11, reducing deployment time by 40%
- Managed 200+ user accounts and permissions via NAS, conducting access audits and boosting system uptime by 15%
- Enhanced endpoint security with CrowdStrike for 50+ devices, resolving 20+ tickets/week with 100% incident resolution rate

PROJECTS

High-Frequency Market Data Processor | C++, Boost Libraries, Custom Memory Pool, Lock-Free Queues [Month] 2026

- Engineered low-latency market data processing engine with custom memory allocators and lock-free data structures, achieving [X] ns average processing time per message (benchmark against system clock)
- Implemented multi-threaded order book reconstruction with NUMA-aware thread pinning and CPU cache optimization, supporting [X]K messages/second throughput (stress test with synthetic data generator)
- Built real-time technical indicator calculations (SMA, EMA, VWAP) using SIMD instructions and template metaprogramming, reducing computation overhead by [X]% (compare with naive implementation)

Scalable Chat Platform | Go, AWS (ECS, RDS, ElastiCache, ALB), WebSockets, Docker [Month] 2026

- Architected cloud-native messaging platform on AWS ECS with auto-scaling policies, PostgreSQL RDS, and Redis ElastiCache, handling [X] concurrent WebSocket connections (load test with Artillery.js)
- Implemented horizontal scaling with Application Load Balancer and sticky sessions, deploying via GitHub Actions CI/CD with blue-green deployment strategy achieving [X]% uptime (CloudWatch metrics)
- Integrated comprehensive observability with CloudWatch dashboards, custom metrics, and structured logging, enabling sub-[X]ms P95 response time monitoring (APM traces and database query analysis)

Real-Time Analytics Pipeline | Apache Kafka, Apache Flink, InfluxDB, React Dashboard, Docker Compose [Month] 2026

- Built event-driven analytics system processing [X]K events/minute through Kafka topics with Flink stream processing, implementing windowed aggregations and complex event pattern detection (producer load testing)
- Designed time-series data storage in InfluxDB with optimized retention policies and continuous queries, supporting [X]-day historical analysis with [X]ms query response times (benchmark with InfluxDB CLI)
- Created real-time React dashboard with WebSocket streaming and D3.js visualizations, displaying live metrics with [X]ms end-to-end latency from event ingestion to UI update (browser performance profiling)

TECHNICAL SKILLS

Systems Languages: C++, C, Go, Rust, Assembly (x86-64)

Performance Optimization: Custom Memory Allocators, Lock-Free Data Structures, NUMA Optimization, SIMD Instructions

Concurrency & Parallelism: Multi-threading, Thread Pinning, CPU Cache Optimization, Lock-Free Queues

Cloud & Infrastructure: AWS (ECS, RDS, ElastiCache, ALB, CloudWatch), Docker, Kubernetes, Load Balancing

Data Processing: Apache Kafka, Apache Flink, InfluxDB, High-Throughput Streaming, Time-Series Databases

Tools & Monitoring: Performance Profiling, Benchmarking, Observability, Load Testing, System Clock Analysis